

SULA, Norway

WMO: 012280

Lat: 63.847N

Lon: 8.467E

Elev: 6

StdP: 101.25

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-5.4	-3.3	-10.2	1.6	-4.0	-8.7	1.8	-2.8	20.9	5.7	19.2	5.7	7.9	110	0.785

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	3.5	19.7	15.4	17.9	14.5	16.6	13.9	16.1	18.5	15.1	17.2	14.3	16.1	4.1	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.1	10.7	16.9	14.2	10.1	16.0	13.4	9.6	15.3	45.1	18.6	42.4	17.3	40.2	16.2	20.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.6	16.3	14.4	DB	-6.0	23.0	2.5	1.8	-7.8	24.3	-9.2	25.3	-10.6	26.3	-12.4	27.6	(4)
(5)			WB	-7.2	17.7	2.3	1.5	-8.8	18.7	-10.1	19.6	-11.4	20.4	-13.1	21.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	7.4	2.7	2.3	3.2	5.4	8.0	10.5	13.0	13.9	12.1	8.5	5.3	3.6
	DBStd	4.75	2.64	2.92	2.54	2.40	2.43	2.06	2.18	1.99	2.11	2.46	2.88	3.06
	HDD10.0	1282	227	214	212	139	71	16	1	0	6	58	141	197
	HDD18.3	3983	486	448	470	387	319	234	165	137	187	305	390	455
	CDD10.0	342	0	0	0	2	11	32	95	122	68	11	1	0
	CDD18.3	2	0	0	0	0	0	0	1	1	0	0	0	0
	CDH23.3	3	0	0	0	0	0	0	3	1	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	6.4	7.7	7.6	7.4	5.8	5.2	5.1	4.5	4.8	6.4	7.0	7.6	8.1
Precipitation	PrecAvg	1457	138	134	136	79	75	97	92	123	150	141	135	156
	PrecMax	1744	348	316	336	183	150	195	175	256	377	261	304	328
	PrecMin	1055	6	6	47	16	40	32	25	37	52	30	5	44
	PrecStd	188	84	69	76	46	27	39	39	54	79	49	89	65
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.5	8.1	9.1	15.1	18.1	19.7	22.7	22.2	18.7	14.9	11.7	9.8
		MCWB	6.9	6.6	6.0	10.1	12.2	14.3	16.8	16.5	14.8	11.4	9.9	8.3
	2%	DB	7.6	7.4	7.8	11.5	14.8	16.7	19.4	20.0	17.0	13.5	10.7	9.1
		MCWB	6.3	6.0	6.0	7.9	10.9	13.2	15.6	16.1	13.7	11.0	9.1	7.8
	5%	DB	7.0	6.8	7.2	9.6	12.9	14.6	17.5	18.1	15.7	12.5	9.8	8.4
		MCWB	5.8	5.4	5.7	7.1	9.8	12.0	14.7	14.9	13.1	10.4	8.1	7.0
	10%	DB	6.3	6.0	6.6	8.3	11.2	13.2	16.0	16.7	14.7	11.6	9.0	7.6
		MCWB	5.1	4.6	5.1	6.4	8.9	11.2	13.9	14.4	12.7	9.8	7.4	6.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.5	7.2	7.2	10.3	13.0	15.5	17.3	17.9	15.6	12.6	10.5	9.0
		MCDB	8.0	7.6	7.8	14.5	16.5	18.4	20.9	20.3	17.4	13.7	11.3	9.5
	2%	WB	6.8	6.5	6.8	8.6	11.3	13.5	16.0	16.4	14.3	11.4	9.5	8.1
		MCDB	7.4	7.2	7.4	10.8	14.4	15.9	18.5	18.7	16.0	12.8	10.4	8.9
	5%	WB	6.0	5.6	6.1	7.6	10.1	12.2	15.0	15.3	13.6	10.7	8.5	7.1
		MCDB	6.8	6.5	7.0	9.1	12.5	14.2	17.2	17.3	15.1	12.0	9.6	8.1
	10%	WB	5.3	4.8	5.2	6.8	9.2	11.4	14.1	14.6	13.0	10.1	7.6	6.2
		MCDB	6.2	5.8	6.3	7.9	11.0	13.0	15.8	16.3	14.4	11.4	8.8	7.4
Mean Daily Temperature Range	5% DB	MDBR	2.8	3.0	3.1	3.5	3.8	3.4	3.6	3.5	3.2	2.7	2.7	2.9
		MCDBR	3.3	3.2	3.7	5.8	6.9	6.0	6.8	6.2	4.4	3.4	3.0	3.4
		MCWBR	3.3	3.3	3.0	3.4	3.8	3.6	3.7	3.3	2.6	2.7	2.9	3.5
	5% WB	MCDBR	3.2	3.2	3.3	5.0	6.2	5.3	6.2	5.4	3.9	3.1	2.9	3.2
		MCWBR	3.3	3.4	3.0	3.3	3.8	3.5	3.8	3.2	2.7	2.8	3.1	3.5
Clear-Sky Solar Irradiance	taub	0.391	0.319	0.341	0.361	0.355	0.356	0.364	0.355	0.342	0.316	0.361	0.360	
	taud	2.600	2.248	2.330	2.373	2.454	2.483	2.488	2.513	2.513	2.393	2.544	2.442	
	Ebn at Noon	313	592	746	817	859	864	845	821	753	602	308	136	
	Edh at Noon	29	59	80	95	96	95	93	83	68	50	26	16	
All-Sky Solar Radiation	RadAvg	0.12	0.58	1.79	3.46	4.71	5.06	4.68	3.59	2.12	0.91	0.21	0.06	
	RadStd	0.02	0.10	0.25	0.29	0.23	0.49	0.42	0.35	0.26	0.13	0.04	0.01	

Historical Trends

		DBAvg	Heating		Cooling		Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	-0.85	N/A	N/A	N/A	N/A	(46)
(47) Regional (28 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page